

Final Project – Spring 2025

Project Summary Report

Fill the document as per the requirements of the project. You must include the following sections in the document:

1. High level description of the organization for which the network plan is being created.
2. Description of end devices.
3. Description of intermediate devices.
4. Screenshots of the completed network.
5. Screenshots demonstrating working of the network in packet tracer.
6. Any other content needed. Refer to the project grading rubric for further details.

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Network Design Project Document

1. High-Level Description of the Organization

This network plan is designed for a **mid-sized pharmacy organization** that operates across multiple locations, including a **Head Office** and **three branch outlets**. The primary goal of this network is to enable seamless communication between the head office and its branches while ensuring local connectivity within each branch for smooth daily operations. The organization provides essential pharmaceutical services to customers and requires a robust and secure network to support internal operations such as inventory management, billing, research, and administrative coordination. Each site has a DHCP server which gives IP addresses to the Devices in the LAN. The Head office also has a DNS and WEB server. The organization has a **web page** that the staff can access connected to a pc from any branch. Each site has the ability to securely exchange data with the head office.

2. Description of End Devices

Each location in the organization's network is equipped with a variety of **end-user devices** that support day-to-day activities. These include **desktop computers (PCs)** for employees to perform administrative tasks, access databases, manage inventory, and interact with customer management systems. Additionally, each branch and the head office are fitted with **network printers**, allowing employees to print invoices, medical records, inventory lists, and other essential documents. These printers are connected to the local area network (LAN), making them accessible from any computer within the same branch. Each site also had a **DHCP server**. Each end device was set to DHCP so that they get the IP, gateway, and DNS address from the DHCP server. The Head office also had a **DNS and WEB server** which enabled other PCs to use the website. The name of the website was saved as record in the DNS server and giving the IP address of the WEB server. The number of PCs per location varies based on the staff size, with the head office hosting more computers due to its administrative functions, while each branch has a smaller, streamlined setup to support front-desk and inventory staff.

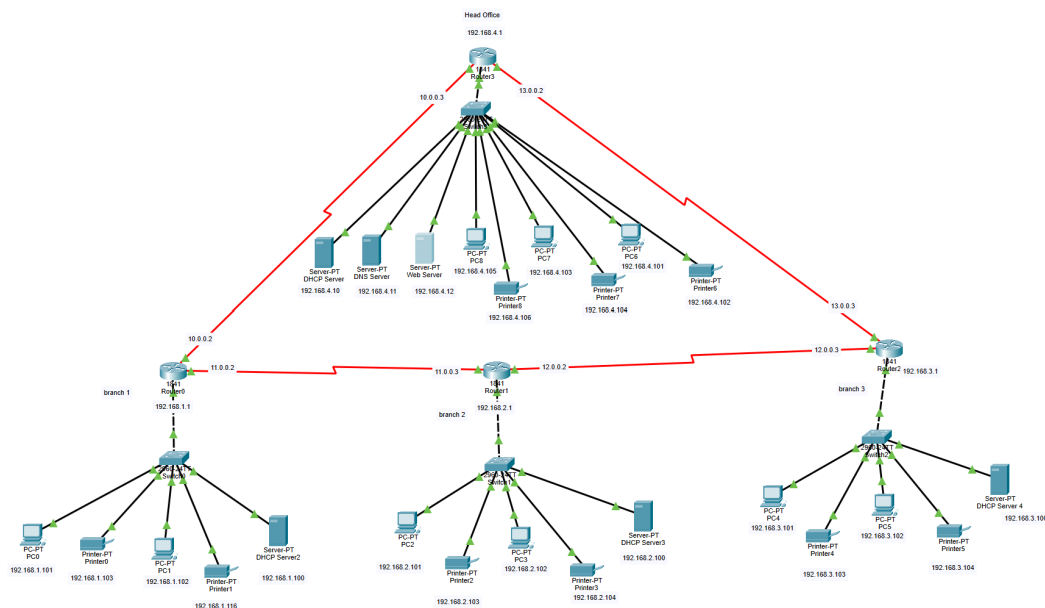
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3. Description of Intermediate Devices

The network is supported by a well-structured combination of **intermediate devices** that manage the flow of data and ensure connectivity between locations. Each site contains a **router** that facilitates communication between branches and provides access to external networks. These routers are dynamically configured with **RIP routing**, offering greater control over the network flow. The routers are connected to each other through the serial ports and these ports are also given a IP address. Connected to each router is a **network switch**, which serves as the central device in the local network, allowing multiple end devices to communicate within the branch. The switches use Ethernet connections to link the computers and printers, forming the internal communication backbone of each site. **Copper Straight-Through cables** are used to connect end devices to the switch, while **Copper Cross-Over cables** are utilized for router-to-router connections, enabling direct communication between routers without intermediary devices. This physical topology ensures that the network is organized, manageable, and easy to troubleshoot.

4. Screenshot of the Completed Network

The completed network layout clearly illustrates the hierarchical structure and interconnectivity between the head office and branch locations. The WAN follows a **ring topology**, where each router is connected to each other and form a closed loop. Each LAN follows a **star topology** with the switch acting as the centre. Each site has its own switch that manages the LAN devices. The structure allows centralized management from the head office while keeping the branches connected and operational.

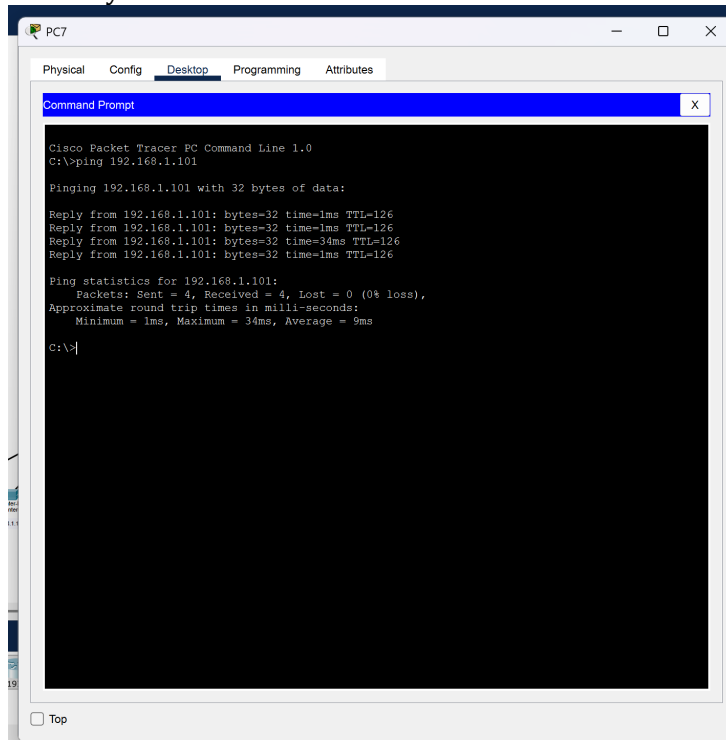


5. Screenshots Demonstrating Working of the Network

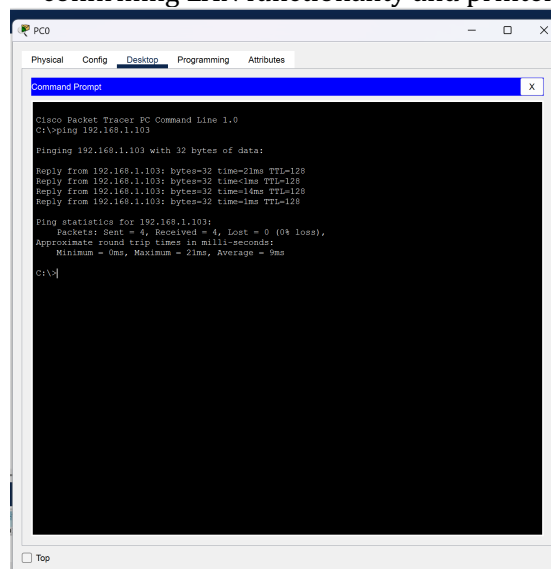
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The following screenshots demonstrate that the network is fully functional in Cisco Packet Tracer. These tests validate end-to-end connectivity, proper IP addressing, and the ability of all devices to communicate effectively across branches.

- **Ping Test Between Branch PCs:** A successful ping from a computer in Head Office to a computer in the branch 1 confirms that routing is configured correctly.

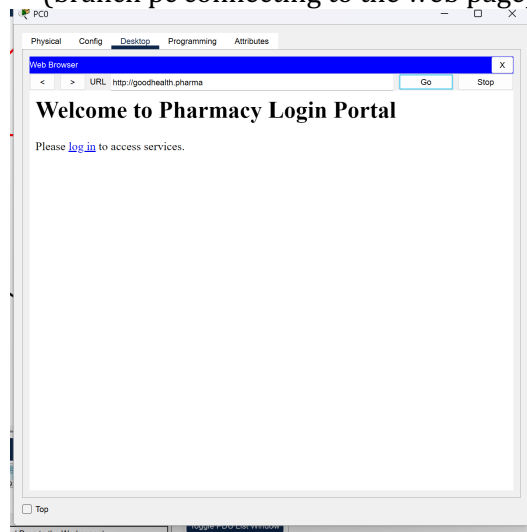


- **Printer Connectivity Test:** A PC successfully pings to the local printer, confirming LAN functionality and printer availability on the network.

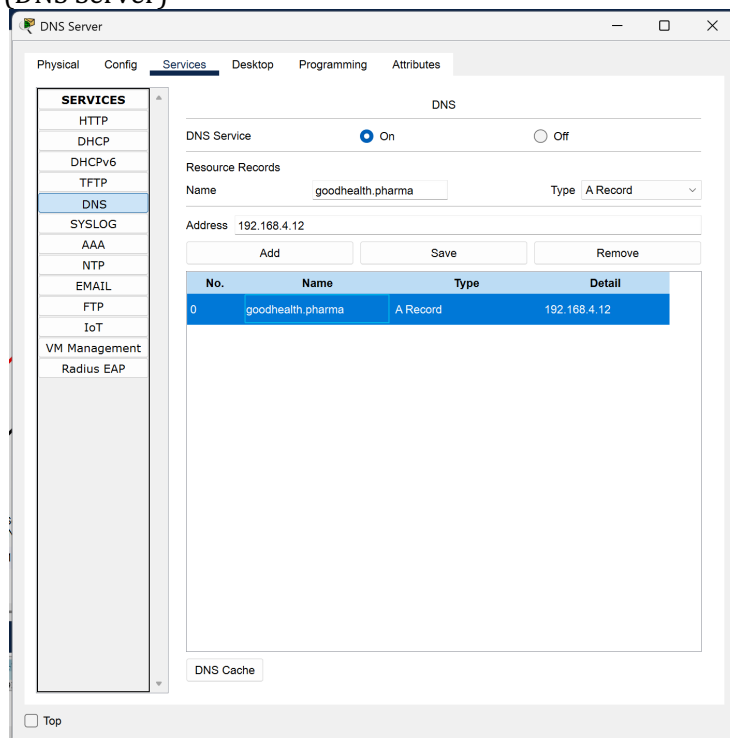


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- **WEB and DNS servers:** Both the head office and the branches can access the website through the web and dns servers placed in the head office.
(branch pc connecting to the web page)

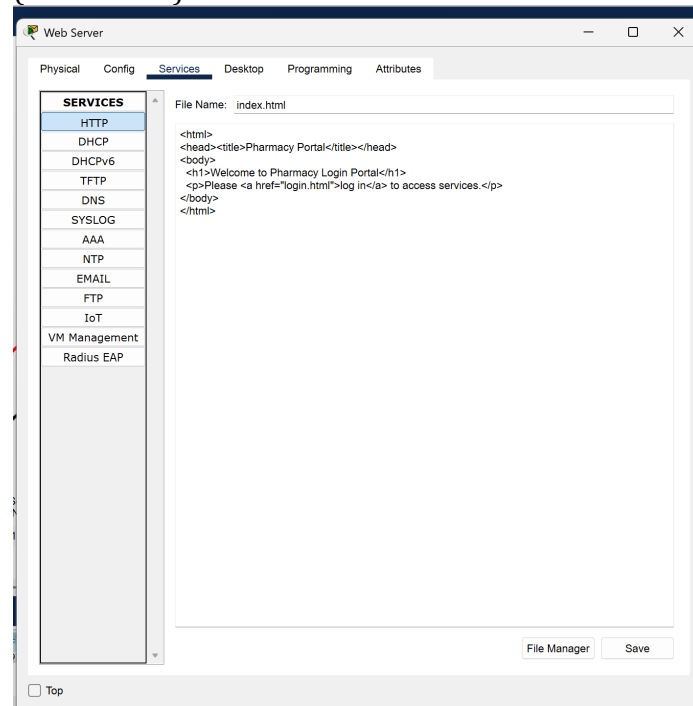


(DNS Server)-



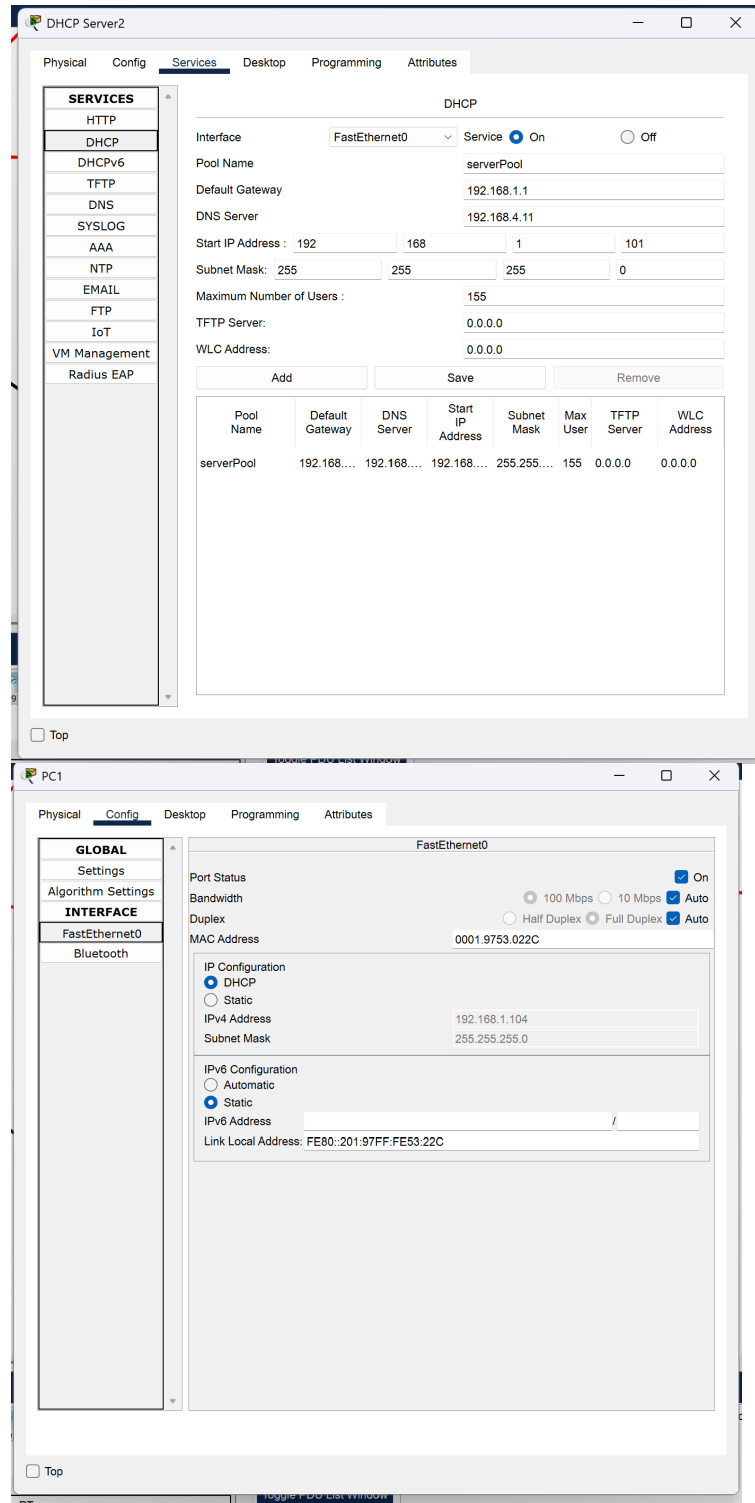
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(WEB server)-



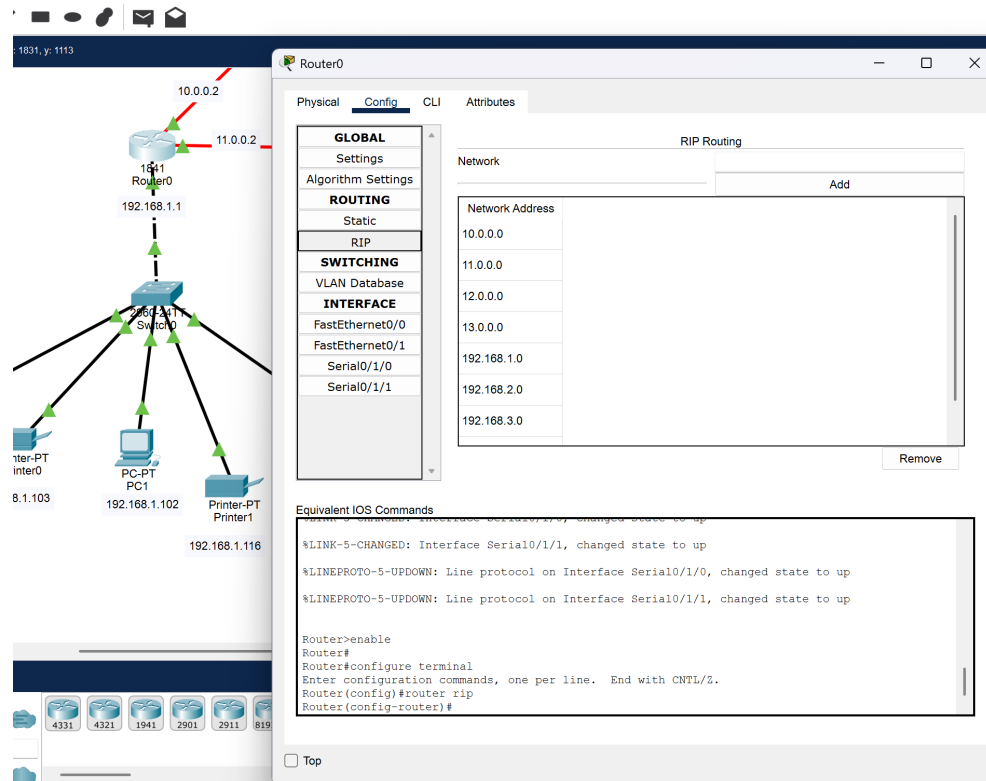
- **DHCP Server:** dedicated DHCP servers in each LAN assign IP addresses and inform about the gateway and DNS addresses to the device connected in the LAN.

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- **RIP routing used:** The routing tables in each router show correctly configured dynamic routes, confirming that each route works correctly.

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6. Additional Content and Project Highlights

This project not only fulfills the basic technical requirements of network connectivity but also applies good design principles to build a system that is maintainable and suited to the organization's operational needs. The use of RIP routing makes sure the connectivity between branches is hard to break because of dynamic routing. Furthermore, consistent IP addressing and device labeling improve manageability and reduce administrative overhead.